

ECON 403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

Fall 2025

Department of Economics
Nazarbayev University

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Lecture time/Location: Section 1L (ECO403): 18:00–19:15, Section 2L (ECO403) & 1L (ECO503): 16:30–17:45. Location TBA, classroom might change.

Office hours: By appointment, book weekly at <https://econ403-503.youcanbook.me>, please sign at least one day in advance.

Course Description

This course analyzes problems created by informational asymmetries and how well-designed market mechanisms—especially auctions—can address them. Auctions are used widely, from art and fresh fish to electricity, government bonds, procurement, and online advertising.

We begin with a short module on the standard auction formats under *complete information*. This serves both as a focused review of game-theoretic tools (strategies, mixed strategies, best responses, dominance, Nash and subgame-perfect equilibrium) and as a baseline for understanding the limitations of complete-information models. Motivated by those limitations, we then develop the formal apparatus for reasoning about knowledge, beliefs, and information, and introduce *Bayesian games* as the central framework for the remainder of the course.

With these tools in place, we study auctions under *incomplete information* (private and interdependent values), derive core results in auction theory (e.g., revenue equivalence and the role of reserve prices), and, if time permits, examine information disclosure and entry. We conclude with an introduction to mechanism design and the theory of optimal auctions. If time permits, we will cover selected advanced topics in auction theory, mechanism design, and information design.

Course Requirements

Official prerequisites. ECON 201 (Intermediate Microeconomics) and ECON 301 (Econometrics I), each with a grade of C- or higher; these are the *minimum, official requirements*.

Expected preparation. You should be proficient in calculus (differentiation and integration, the Fundamental Theorem of Calculus, iterated integrals, unconstrained and constrained optimization), probability and statistics (random variables, distributions, expectation and conditional expectation, transformations of random variables, independence), and mathematical reasoning (reading notation, following and producing proofs). A background exceeding a single Calculus I course is strongly recommended.

Unofficial prerequisite (Game Theory). Prior exposure at the level of ECON 302 or similar.¹ At a minimum, you should know normal-form games, strategies and expected utility, dominance and rationalizability, Nash equilibrium (including mixed strategies), extensive-form games, and subgame-perfect equilibrium.

Course Aims

1. Understand the basic framework to model information and knowledge in strategic situations.
2. Analyze the main auction formats and the seminal results in auction theory.
3. Gain in-depth knowledge into the techniques used in modern microeconomic analysis.
4. Use the tools learned in the course to be able to model phenomena observed in real life.

¹ECON 305: Advanced Microeconomics is also suitable. A widely used course on Coursera covering similar material may serve as a substitute for ECON 302.

Course Learning Outcomes

By the end of the course the student will be expected to be able to

1. Demonstrate competence in using formal tools to model information in strategic situations.
2. Analyze games of incomplete information utilizing standard equilibrium concepts.
3. Demonstrate knowledge of the standard auction formats and contrast them in terms of their equilibrium outcomes.
4. Propose, construct and solve a simple mathematical model to analyze a phenomenon from real life using the tools from the course.

Learning and Teaching Methods

Lectures, problem sets, and a final essay.

Course Textbooks

There is no textbook required for this course. The course is based mainly on my lectures, slides and readings. All materials will be uploaded to Moodle.

The following is a list of undergraduate and graduate textbooks that you may find useful:

- Maschler, M., Solan, E., & Zamir, S. (2013). *Game Theory*. Cambridge: Cambridge University Press.
- Milgrom (2003), *Putting Auction Theory to Work*, Cambridge University Press.
- Krishna (2009), *Auction Theory*, Academic Press.

You will be responsible for all the material I cover or assign in class or tutorials. Therefore, the textbooks should be seen as a complement to the lectures are definitively *not a perfect substitute* for them.

Course Website

Additional course material, such as lecture slides, problem sets, and announcements will be posted on the Moodle page of the course, which you should check regularly.

Assessments

Table 1: Assessment weights

ECO403 (Undergraduate)	ECO503 (Master's)
Problem Sets (15%)	Problem Sets (15%)
Essay (10%)	Research Proposal (20%)
Midterm (30%)	Midterm (25%)
Final Exam (35%)	Final Exam (30%)
Post-lecture Quizzes (5%)	Post-lecture Quizzes (5%)
Midsemester Project (5%)	Midsemester Project (5%)

Problem Sets There will be at least five problem sets. They will be assigned in class and are due *one week after assignment*. Late submissions will **not** be accepted. Problem sets are graded on a 0–2 scale:

- 0: Not submitted.
- 1: Submitted but incomplete.
- 2: Submitted and complete.

Solutions will be discussed in class or tutorials. *Students are responsible for comparing their work with the discussion to verify correctness and identify mistakes.* Typeset submissions are *encouraged but not required* (LaTeX preferred; Word acceptable). **All submissions—whether handwritten or typeset—must be uploaded to Moodle as a single PDF file; paper submissions will not be accepted.**

Midsemester Project A substantial, exam-style problem set requiring a written report/solution. The report must be typeset and submitted as a PDF (LaTeX strongly preferred; Word acceptable). **Handwritten reports will not be accepted.** The project is intended to mirror the scope and level of the final exam. Unless otherwise announced, this is individual work. A brief rubric will be provided in advance; evaluation will emphasize correctness and rigor of analysis, clarity of exposition, and presentation quality.

Essay (ECO403) Undergraduate students submit a final essay proposing a simple microeconomic model of a real phenomenon, analyzing and solving it, and communicating findings clearly. Assessment will consider model appropriateness, depth and correctness of analysis, and quality of writing.

Research Proposal (ECO503) Master's students submit a research proposal on a topic related to the course. Students are encouraged to consult the instructor to refine the topic. Detailed guidelines will be provided during the course.

Midterm The midterm will take place in *Week 6* and covers approximately the first half of the course. The exact date will be announced during the term.

Final Exam The final examination covers the second half of the course. The date is set by the Registrar during *Finals Week*. Further information on its format and permitted materials will be provided closer to the date; its level of rigour will be comparable to the Midsemester Project.

Post-lecture Quizzes After each lecture, a brief Moodle quiz (typically a single true/false question) will be available. The deadline is 17:50 on the day of the next lecture.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above.

Grading

Your course grade is determined by the weighted average of the assessments above according to the following scale:

A	95–100	A–	90–94.99
B+	85–89.99	B	80–84.99
B–	75–79.99	C+	70–74.99
C	65–69.99	C–	60–64.99
D+	55–59.99	D	50–54.99
F	0–44.99		

Course Expectations

1. Class will begin promptly at the scheduled starting time. As a courtesy to others, it is requested that you not arrive after class has begun or leave before it ends. PLEASE NOTE: Since showing up late and leaving early is disruptive to the class, students who disrupt the class in this or any other manner may be administratively dropped or have their course grade reduced at the instructor's discretion. If you must leave class early or will be late, please let me know before class begins.
2. Class attendance will be taken regularly and randomly throughout the semester. If you fail to attend a lecture, you must submit the documents justifying your absence to the registrar and notify me by email. Students who have more than 25% of unexcused absences (7 lectures) will not be able to take the final exam. The instructor holds the right to excuse students from class or administratively drop them if necessary for violation of this policy. Failure to attend lectures during the add/drop period will result in your dismissal from the course.
3. *Exams* will be held outside of class, typically on Fridays or weekends. The midterm exam is closed book and closed notes. The use of programmable calculators is prohibited. The dates for the midterm exam is fixed, but their locations will be announced later. The final exam will be scheduled by the Office of the Registrar.

If you miss an exam, you will receive a grade of zero unless you provide a valid and documented reason. In the case of a missed midterm exam with valid documentation, you will be allowed to take a make-up exam, which will be conducted as an oral exam. If you miss the final exam for a valid and documented reason, the University's Grading Policy regarding the grade of Incomplete will apply.

The final exam is cumulative. Appeals for midterm exam grades must be submitted in writing within five days of receiving the graded exam. Appeals for the final exam grade or the final course grade must also be submitted in writing within two days of the grades being posted.

4. The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available.²

Academic Misconduct

Students are expected to act with integrity. Cheating and plagiarism will not be tolerated. Any student deemed academically dishonest will be reported to the SSH disciplinary committee. Any cases of academic misconduct will be handled in accordance with the procedures of the Student Code of Conduct. The penalties may include anything from failing grade on a particular assignment to a loss of stipend, and even, dismissal from the University. You can read more about academic misconduct in the Student Code of Conduct. https://aao.nu.edu.kz/academic_integrity

AI Policy

This policy applies to any generative AI tool and all AI-generated products. You are discouraged from using AI tools unless explicitly instructed by me. If you are unsure or have questions, you must contact me before using AI for any assignment.

In this course, the potential risks of AI usage generally outweigh the benefits to student learning. If AI use is permitted, you are required to clearly indicate which parts of your assignment were created using AI and which parts are your original work. Additionally, no more than 25% of an assignment may be generated with AI, even when its use is approved.

You are required to sign an AI usage contract confirming your understanding and agreement to these policies. Unacknowledged use of AI tools will be treated as plagiarism, and penalties will be applied in accordance with the NU Student Code of Conduct.

Tentative Course Outline

Session	Topics and Assignments	Course Aims	CLOs
Week 1	Introduction/Knowledge	1,3	1
Week 2	Auctions with complete information	1,2,3,4	1,2,3
Week 3	Knowledge/Beliefs	1,3	1
Week 4	Beliefs/Bayesian Games	1,3	1,2
Week 5	Bayesian Games	1,2,3,4	1,2
Week 6	Midterm 1		
Week 7	Bayesian Auctions, Private values	1,2,3	1,2,3
Week 8	Fall Break		
Week 9	Private values, Revenue Equivalence	1,2,3,4	1,2,3
Week 10	Extensions: asymmetries, risk aversion, reserve prices	1,2,3,4	1,2,3
Week 11	Introduction to Mechanism Design	1,2,3,4	1,2,3
Week 12	Optimal Auction Design	1,2,3,4	1,2,3
Week 13	Interdependent Value auctions	1,2,3,4	1,2,3
Week 14	Advanced Topics: revenue linkage, information	1,2,3,4	1,2,3
Week 15	Advanced Topics: Information Design	1,2,3,4	1,2,3

Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.

²For more information, please contact us at SLNC@nu.edu.kz and follow this link to the Accommodation Request Instructions.